

## Practice Midterm

**Problem 1.** Consider generating a random variable  $X$  such that

$$\mathbb{P}(X = i) = \frac{e^{-\lambda} \lambda^i / i!}{\sum_{j=0}^k e^{-\lambda} \lambda^j / j!}, \quad i = 0, 1, 2, \dots, k.$$

- (a) Derive the inverse transform method to generate  $X$ .  
 (b) Suppose you run the following algorithm to generate  $X$ : First sample from a Poisson distribution with rate  $\lambda$ , and if this value falls into  $\{0, 1, \dots, k\}$ , you output it; otherwise you return to the sampling step and repeat until you obtain an output. Is this an acceptance-rejection procedure? If it is, state precisely all the algorithmic configurations in this procedure.

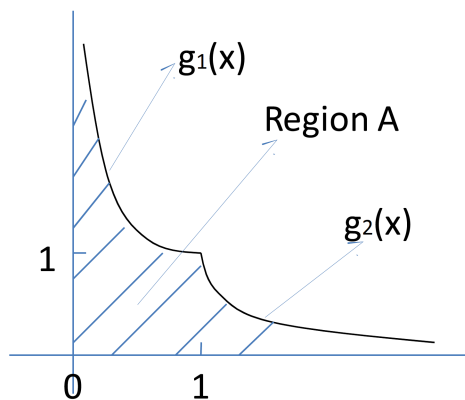
**Problem 2.** Let  $\alpha, \beta > 0$ . Give a method for generating a random variable having distribution function

$$F(x) = 1 - \exp(-\alpha x^\beta), 0 < x < \infty.$$

- (a) Derive an inverse transform method to generate  $X$ .  
 (b) Suppose you insist to use an acceptance-rejection method to generate  $X$ , using a proposal that has the same distribution function as above but with a new  $\tilde{\beta}$  such that  $\tilde{\beta} > \beta$ , and that your computer has an established routine to generate from this proposal distribution directly. Does this acceptance-rejection method work? Explain.

**Problem 3.** Provide a method to sample uniform points in the region  $A$  under the curve  $g_1(x)$  and  $g_2(x)$  and the axis as indicated in the picture. Given  $\alpha \in (0, 1)$ , the curves  $g_1(x)$  and  $g_2(x)$  are described by

$$g_1(x) = x^{\alpha-1} \mathbb{1}\{0 < x \leq 1\}, \quad g_2(x) = x^{1/(\alpha-1)} \mathbb{1}\{1 \leq x < \infty\}.$$



**Problem 4.** Consider the continuous nonnegative random variable  $X$  with probability density function

$$f_X(x) = \int_0^\infty \frac{9}{(1+xt)^4} t^3 e^{-t^3} dt, \quad x \geq 0.$$

Derive an algorithm to generate  $X$ .